Online Resource 1 to "Confinement-loss prediction in photonic crystal fiber SPR sensors: kernel surrogates versus generative data augmentation"

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Abstract

This supplement contains material supporting the main article that is not required to follow its argument: the dual formulation of support vector regression and its optimality conditions (Sect. S1), the Bayesian hyperparameter search space and its outcome (Sect. S2), per-fold leave-one-geometry-out results (Sect. S3), the generative model configuration and the contradiction-search protocol (Sect. S4), a Gaussian process kernel and regulariser ablation (Sect. S5), and the corrected Monte Carlo tolerance tables (Sect. S6), and the predicted spectra underlying the sensitivity quoted in the article (Sect. S7). Section, table, figure and equation numbers are prefixed with S; citations refer to the reference list of the main article, reproduced here for convenience.

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S1 Support vector regression: dual formulation

The main article states the primal objective in Sect. 4.3. The derivation below is the standard one [4], reproduced here so that the argument of Sect. 5.2 of the main article can be followed without a second source. Introducing Lagrange multipliers $\alpha_i, \alpha_i^* \geq 0$

for the two ϵ-tube constraints and $\eta_i, \eta_i^* \geq 0$ for the non-negativity of the slacks gives the Lagrangian

$$\begin{aligned} L = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) - \sum_{i=1}^{n}(\eta_i\xi_i + \eta_i^*\xi_i^*) \\ - \sum_{i=1}^{n}\alpha_i(\epsilon + \xi_i - y_i + \langle w, x_i\rangle + b) \\ - \sum_{i=1}^{n}\alpha_i^*(\epsilon + \xi_i^* + y_i - \langle w, x_i\rangle - b)\,. \end{aligned} \tag{S1}$$

Setting the partial derivatives with respect to the primal variables to zero yields the saddle-point conditions

$$w = \sum_{i=1}^{n}(\alpha_i - \alpha_i^*)x_i, \qquad \sum_{i=1}^{n}(\alpha_i - \alpha_i^*) = 0, \qquad \alpha_i^{(*)} + \eta_i^{(*)} = C, \tag{S2}$$

and substituting them back removes the primal variables entirely, leaving the dual

$$\begin{aligned} \max_{\alpha,\alpha^*} \quad & -\frac{1}{2}\sum_{i,j=1}^{n}(\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*)K(x_i, x_j) \\ & -\epsilon\sum_{i=1}^{n}(\alpha_i + \alpha_i^*) + \sum_{i=1}^{n}y_i(\alpha_i - \alpha_i^*), \end{aligned} \tag{S3}$$

subject to $\sum_i(\alpha_i - \alpha_i^*) = 0$ and $\alpha_i, \alpha_i^* \in [0, C]$. The prediction is then

$$f(x) = \sum_{i=1}^{n}(\alpha_i - \alpha_i^*)K(x_i, x) + b. \tag{S4}$$

Two properties of (S3)–(S4) matter for the argument in the main article. First, the inputs enter only through the kernel $K(x_i, x_j)$, so the RBF map never has to be formed explicitly. Second, the Karush-Kuhn-Tucker complementarity conditions

$$\alpha_i(\epsilon + \xi_i - y_i + \langle w, x_i\rangle + b) = 0, \qquad (C - \alpha_i)\xi_i = 0, \tag{S5}$$

force $\alpha_i = \alpha_i^* = 0$ for every sample strictly inside the ϵ-tube. The expansion in (S4) is therefore supported only by samples on or outside the tube boundary. On this dataset that concentrates the fit near the resonance transition, where the response is steepest, which accounts for the model's tolerance of a 432-sample training set.

Condition (S5) also explains the contrast with Gaussian process regression [3] in Sect. 5.2 of the main article. A pair of near-duplicate inputs carrying contradictory targets saturates at $\alpha_i = C$: both points become bounded support vectors and the fitted function is degraded, but nothing in (S3) becomes ill-posed, since no matrix inversion is involved.

S2 Bayesian optimisation of the SVR hyperparameters

Table S1 gives the search space and the selected configuration. The acquisition function is expected improvement over a Gaussian process surrogate on the hyperparameter space [5], with 32 evaluations. Each candidate is scored by three-fold cross-validated mean squared error computed on the training partition only, so no information from the held-out geometry enters the selection.

Table S1 Hyperparameter search space and the configuration selected by Bayesian optimisation. Priors are log-uniform because the three parameters span several orders of magnitude and the objective is far more sensitive to their order than to their mantissa.

Parameter	Range	Prior	Selected
C (box constraint)	$[1, 2000]$	log-uniform	2000
γ (RBF width)	$[10^{-4}, 1]$	log-uniform	0.0172
ϵ (tube)	$[10^{-4}, 10^{-1}]$	log-uniform	10^{-4}
Kernel	RBF (fixed)	—	RBF
Evaluations	32	—	—
Inner validation	3-fold CV	—	—

The selected C and ϵ sit at their respective range limits, which indicates the optimum lies at or beyond the boundary: the fit prefers a hard penalty and an almost vanishing insensitivity tube. Widening the range is unlikely to change the fit materially – at $\epsilon = 10^{-4}$ the tube is already far narrower than the residual scale – but the boundary solution should be reported rather than presented as an interior optimum.

The selected width $\gamma = 0.0172$ is small relative to the unit-cube extent of the scaled inputs, giving a broad kernel: the model is smoothing over a substantial fraction of the design space rather than fitting locally, which is the expected regime when nine geometries must support prediction on a tenth.

S3 Per-fold leave-one-geometry-out results

Table S2 gives the fold-level results summarised in Table 2 of the main article, and Fig. S1 shows them graphically.

The per-fold view supports a conclusion the averages cannot. Augmentation is not a uniform penalty: it is nearly neutral on D3 and D4, beneficial on D5, and severe on D8, where the error grows by a factor of 23. This is the signature of synthetic samples that contradict specific regions of the design space, rather than of noise added evenly across it. D5 is the only fold on which augmentation helps, and it is also the fold with the highest real-only error, which is consistent with additional samples helping only where real coverage was genuinely sparse.

Table S2 Held-out mean squared error for each of the nine geometric configurations, with and without GAN-generated training samples. Folds run down the rows; the final two rows give the summary statistics quoted in the main article.

Held-out geometry	SVR, real data only	SVR + GAN samples	Ratio
D1	0.128	1.074	8.41
D2	0.379	0.848	2.24
D3	0.257	0.321	1.25
D4	0.212	0.261	1.23
D5	0.829	0.487	0.59
D6	0.121	1.359	11.27
D7	0.131	0.684	5.21
D8	0.096	2.200	22.83
D9	0.060	0.194	3.24
Mean	0.246	0.825	—
Standard deviation	0.226	0.610	—

Ratio is augmented MSE divided by real-only MSE; values above one indicate augmentation made that fold worse.

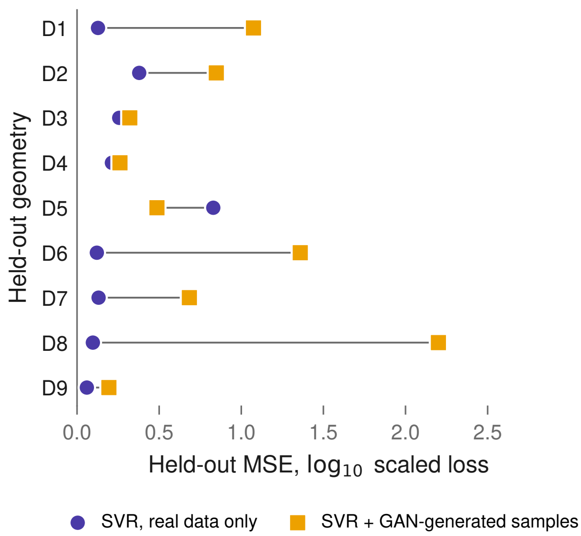

Fig. S1 Per-fold held-out error with and without GAN augmentation. Each connector spans one geometry, so the direction of change is read directly. Eight of the nine folds move to the right, that is, get worse under augmentation.

S4 Generative model and contradiction search

S4.1 WGAN-GP configuration

The generative model is a Wasserstein GAN with gradient penalty [1], reimplementing the architecture reported by Zelaci et al. [6] so that the augmentation experiment tests their procedure and not a variant of it. Table S3 gives the configuration.

Table S3 WGAN-GP configuration used for the augmentation experiments. Generator and critic are given side by side; the training column applies to both.

Property	Generator	Critic	Training
Hidden layers	5	5	—
Units per layer	128	32	—
Activation	LeakyReLU (0.2)	LeakyReLU (0.2)	—
Normalisation	none	batch normalisation	—
Output width	8	1	—
Optimiser	—	—	Adam, $\beta = (0.5, 0.9)$
Learning rate	—	—	10^{-4}
Batch size	—	—	32
Critic steps	—	—	5 per generator step
Epochs	—	—	2000

The generator is trained on the concatenation of the min-max scaled input features and the min-max scaled target, so it models the joint distribution of geometry and response. This is the property the contradiction search exploits: nothing in the objective requires the generated joint samples to be consistent with a single-valued map from geometry to loss.

S4.2 Contradiction search protocol

We drew 50,000 samples from the trained generator and, for each, found its nearest neighbour in the scaled feature space under the Euclidean metric, excluding the sample itself. For a pair at feature distance δ with targets differing by Δy, we ranked by

$$\rho = \frac{\Delta y}{\max(\delta,\, 10^{-6})}, \tag{S6}$$

which measures how steeply the generated response surface must rise to accommodate the pair. Table 4 of the main article reports the three pairs with the largest ρ. Ranking by ρ rather than by δ alone is deliberate: a pair that is merely close is unremarkable, and it is the combination of proximity and target disagreement that is physically impossible.

The feature space for this search is the seven-dimensional set the generator was trained on – analyte index, real effective index, wavelength, pitch and the three hole diameters – min-max scaled to the unit cube. The floor of 10^{-6} in (S6) guards the division for coincident samples.

Note that ρ is a diagnostic, not an estimator: its absolute magnitude depends on the scaling of both axes and should not be compared across datasets. Within this experiment it is the ordering that carries the information.

S5 Gaussian process kernel and regulariser ablation

Section 5.2 of the main article attributes the GPR collapse to ill-conditioning of $[K + \sigma_n^2 I]$. If that account is right, the collapse should depend on how the inversion is regularised. Table S4 tests this by crossing the two available regularisers – an additive white-noise kernel term (WK) [3] and `scikit-learn`'s explicit jitter α – against real and augmented training data.

Table S4 GPR held-out MSE under each combination of regulariser, on real and on augmented training data. Regulariser configurations run down the rows, training sets across the columns.

Regulariser configuration	Real data	GAN-augmented
White kernel only ($\alpha = 0$)	0.185	69.873
White kernel and α	2.116	68.262
α only, no white kernel	71.849	69.873
Neither	—	—

Dashes mark configurations for which no value was recorded. With neither regulariser present, the covariance matrix is singular by construction once two training inputs coincide, so the configuration is not expected to produce a finite result.

On real data the regulariser choice matters enormously: the white kernel alone gives 0.185, while jitter alone gives 71.849 – a 388-fold difference produced by nothing but how the diagonal is stabilised. This is already evidence that the inversion is close to the edge of conditioning even before any synthetic data is introduced.

On augmented data the picture inverts: every configuration lands near 69, regardless of regulariser. Once the contradictory pairs of Sect. S4 are present, no choice of regularisation recovers a usable model. The problem is no longer the conditioning of an otherwise well-posed system, but a training set requiring two different outputs at effectively the same input. Regularisation can accommodate noise; it cannot accommodate contradiction.

S6 Monte Carlo tolerance analysis

S6.1 Protocol

The baseline is the first sampled configuration of the dataset ($\Lambda = 0.24$, $d_1 = 0.25$, $d_2 = 0.75$, $d_3 = 0.35$ in dataset column units) at analyte index 1.33. For each tolerance level $\sigma \in \{1\%, 3\%, 5\%\}$ we drew 100 perturbed geometries, adding to each geometric variable v a Gaussian perturbation of standard deviation σv, and located the predicted resonance as the argmax of the surrogate prediction over a 500-point wavelength grid spanning 500–800 nm. Any trial whose argmax fell in the outermost 2% of the grid was discarded as a boundary artefact rather than a resonance. The reported jitter is the standard deviation of the surviving peak positions. Per-variable volatilities were obtained by repeating the procedure at $\sigma = 3\%$ with one variable perturbed at a time.

S6.2 Correction to the published values

These values supersede those in [2]. The published run evaluated at $\Lambda = 2.0$, $d_1 = 0.225$, $d_2 = 0.375$, $d_3 = 0.175$. Three of those four lie outside the sampled design space, which spans $\Lambda \in [0.15, 0.24]$, $d_1 \in [0.25, 0.45]$ and $d_2 \in [0.55, 0.75]$: the values 0.225, 0.375 and 0.175 are the hole *radii* r_1, r_2, r_3 of Fig. 1 of the main article entered into *diameter* columns, and 2.0 is the lattice pitch in micrometres entered into a column whose own unit is ten micrometres.

The surrogate was therefore extrapolating well outside its training envelope, where it has no interior resonance to find, and the unguarded argmax returned grid endpoints in roughly half of all trials. That is what produced the published peak spread of 2.96 against a 3.00-wide grid. Re-evaluated at a sampled geometry the degeneracy disappears entirely: no trial is rejected at any tolerance level. The qualitative conclusion is unchanged and in fact sharpens, the pitch accounting for 98.8% of the variance in peak position against 95.1% before; the magnitudes fall by roughly a factor of eighteen.

S6.3 Results

Table S5 reports both panels. The upper panel gives the scatter when all four geometric variables are perturbed together, and the lower panel attributes that scatter to individual variables.

Table S5 Monte Carlo peak scatter at each tolerance level, and the per-variable decomposition at 3%, evaluated at a sampled geometry with a boundary-guarded peak search. All values are standard deviations of the predicted resonance position in nanometres.

All variables perturbed simultaneously			
Tolerance level	1%	3%	5%
Peak jitter (std)	2.32 nm	7.18 nm	10.11 nm
Trials rejected	0/100	0/100	0/100
One variable at a time, 3% tolerance			
Variable	Jitter (std)	Share of variance	Rank
Λ	5.49 nm	98.8%	1
d_1	0.43 nm	0.6%	2
d_2	0.38 nm	0.5%	3
d_3	0.22 nm	0.2%	4

Shares are of total variance, $\sigma_i^2 / \sum_j \sigma_j^2$, since variances rather than standard deviations are additive. The decomposition ignores interaction between variables, which is why it need not reproduce the simultaneous-perturbation jitter of the upper panel. The three hole diameters are separated by less than the Monte Carlo error at 100 trials, so their relative ranking should not be read as significant.

Figure S2 shows the per-variable decomposition of the lower panel of Table S5 graphically, on a logarithmic scale.

S7 Spectral validation

Aggregate error statistics can conceal a surrogate that is accurate on average yet incorrect in spectral shape. As a physical check, the trained SVR was swept across the full wavelength range at fixed geometry for each analyte and the predicted spectra examined directly (Fig. S3).

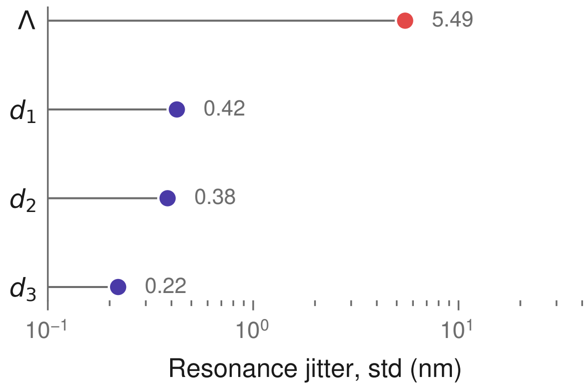

Fig. S2 Resonance jitter attributable to each geometric variable at 3% tolerance, on a logarithmic scale. The lattice pitch (highlighted) exceeds the largest hole-diameter contribution by more than an order of magnitude. Evaluated at a sampled geometry with a boundary-guarded peak search.

The surrogate reproduces the expected behaviour: a single well-formed resonance shifting monotonically to longer wavelengths as the analyte index rises from 1.33 to 1.35, in near-uniform steps of 0.279 and 0.273 dataset units. The phase-matching condition was never supplied to the model, so recovery of both the direction and the regularity of this shift indicates that the underlying coupling has been captured rather than the training spectra memorised. The corresponding sensitivity is 27.63 per refractive index unit in the units of the source dataset.

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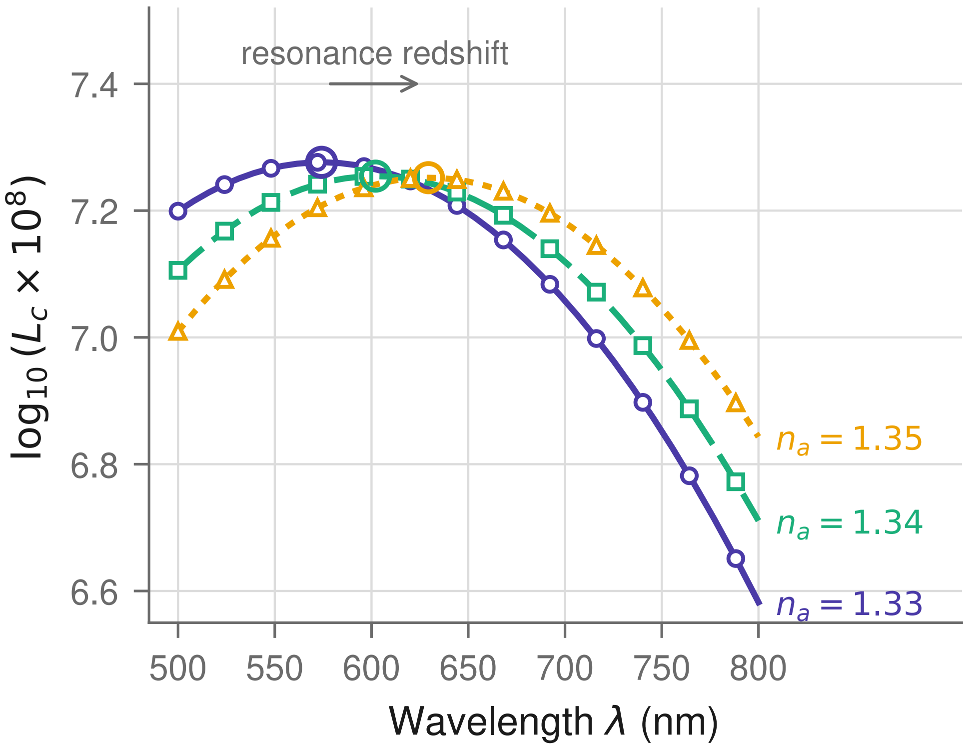

Fig. S3 Predicted confinement-loss spectra at fixed geometry for the three analyte indices. Rings mark the predicted resonance. The peak moves monotonically to longer wavelengths as n_a increases, in uniform steps of 0.276 dataset units per 0.01 refractive index unit.

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